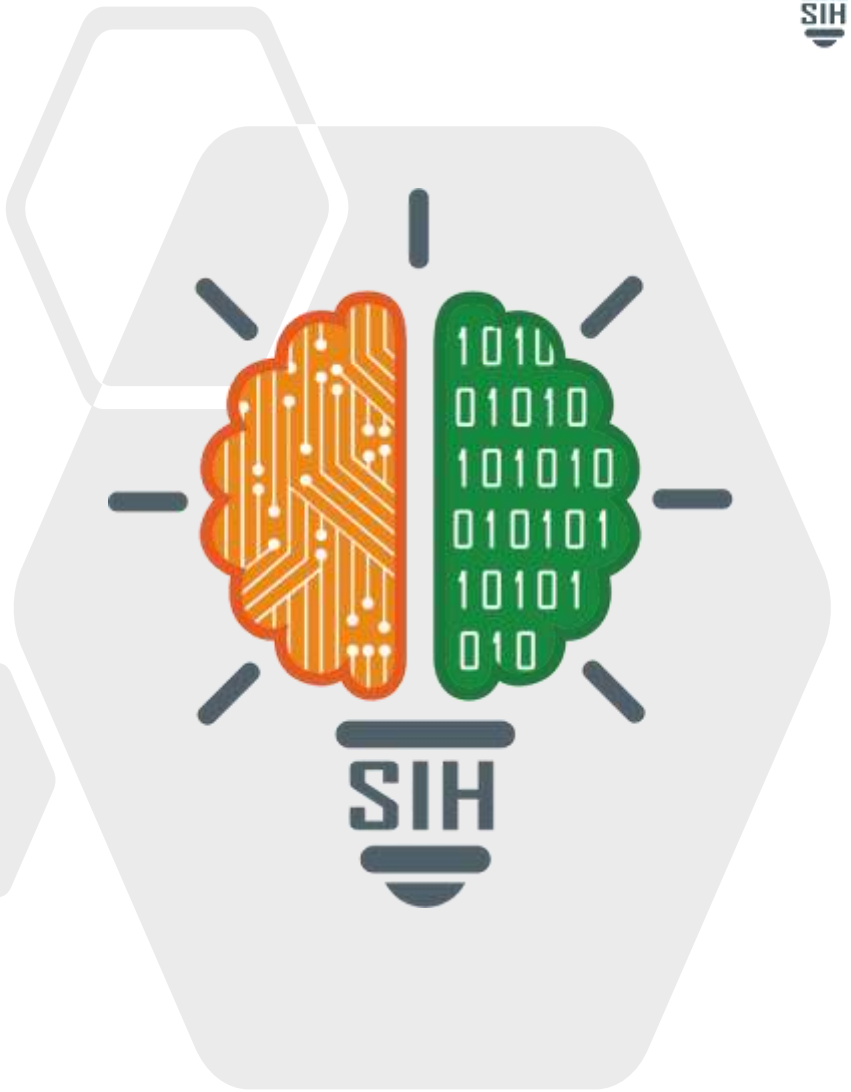
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- **Problem Statement ID – 26155**
- **Problem Statement Title- AI-Driven Multi-Vendor Network Security Compliance Auditor**
- **Theme- Blockchain & Cybersecurity**
- **PS Category- Software**
- **Team ID- 172100**
- **Team Name- KT@UnseenGeeks**



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IDEA TITLE

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Problem :

Heterogeneous networks use different vendor CLI, configuration structures and OS versions, making unified compliance auditing difficult to scale.

Manual/vendor-locked audits limit multi-framework checks, evidence generation and adaptation to new devices.

Our Idea: NOVAIR = Network-Oriented Vendor-Agnostic Intelligence & Risk

Normalize multi-vendor configurations into one Canonical IR.

Run compliance, CVE, PQC and analytics checks in parallel to generate evidence-backed findings and remediation.

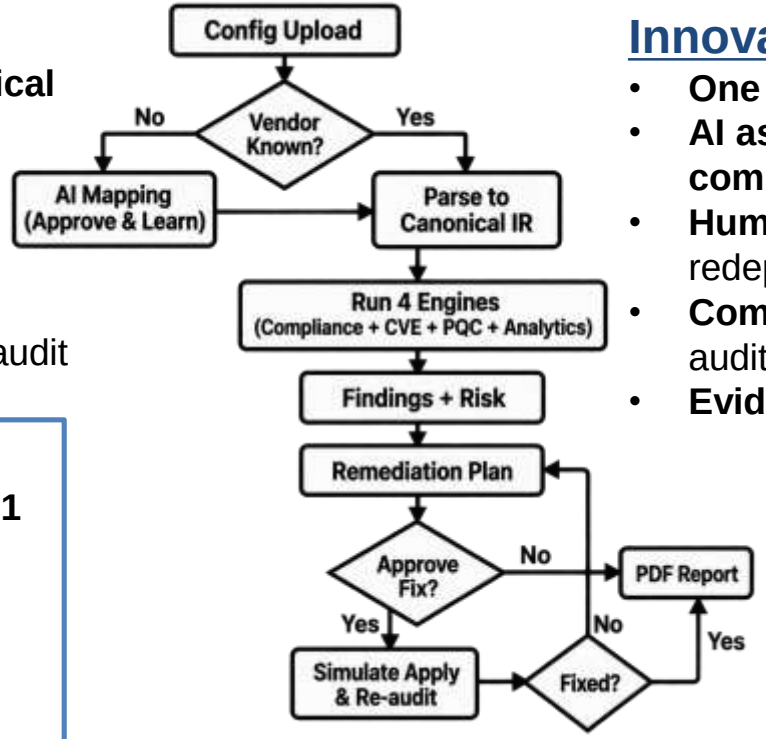
AI-assisted human approval learns unknown syntax for future audits without redeployment.

Proposed Solution :

- **Normalize:** Ingest single/bulk configs → convert to **Canonical IR**
- **Analyze:** CIS + NIST + STIG + ISO + CERT-In
- CVE + PQC + Analytics
- **Adapt:** AI-assisted mapping of unknown syntax → **human approval** → **reuse**
- **Remediate & Verify:** Device-specific fix → simulate → re-audit → **PDF evidence**

Quantified Impact:

- Vendor Normalization: 6 vendor formats consolidated into **1 Canonical IR** with **103 properties + 82 controls**
- Unified Compliance: CIS + NIST + STIG + ISO + CERT-In evaluated in **a single audit run**
- Remediation Verification: Simulated fix reduced findings from **12 to 5**, with **7 verified closures**



Innovation/Uniqueness :

- **One Canonical IR** for multiple vendors
- **AI assists; deterministic rules decide compliance**
- **Human-approved learning** without backend redeployment
- **Compliance + CVE + PQC + Analytics** in one audit
- **Evidence-backed remediation & re-audit**

Target Users:

- Government & PSU security teams
- Network/SOC administrators
- Compliance auditors
- Large enterprise network teams

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TECHNICAL APPROACH

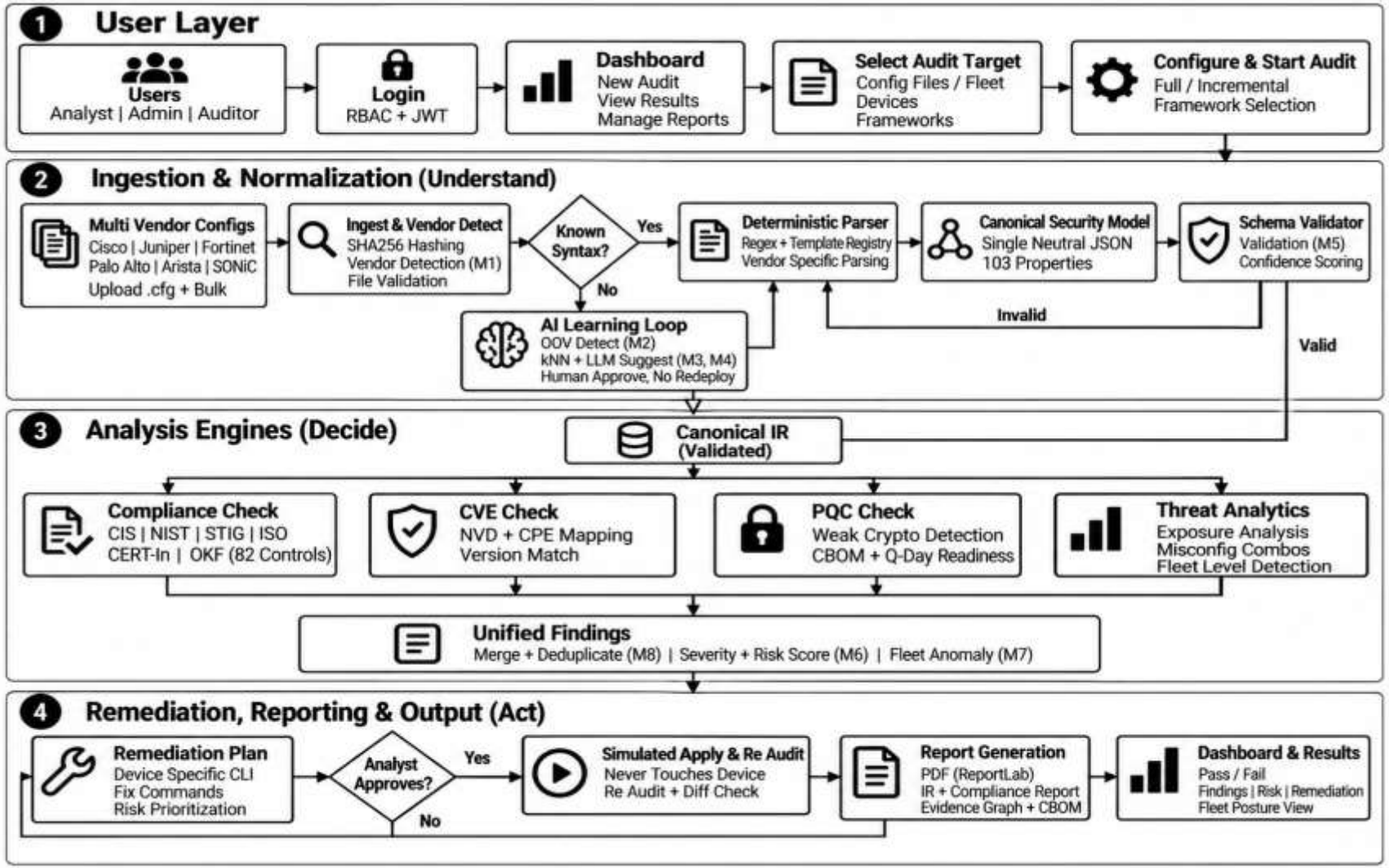
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Core Technologies:

- **Frontend:** React
- **Backend:** FastAPI
- **AI/ML:** MiniLM, kNN, XGBoost, HDBSCAN, OKF
- **Compliance:** YAML, Rule Engine, OSCAL
- **Security:** NVD, CPE, CBOM
- **Deployment:** Docker, On-Prem / Offline

Data Sources:

- **Configs:** Cisco, Juniper, Fortinet, Palo Alto, Arista, SONiC, cloud SG (.cfg/.conf)
- **Frameworks:** CIS, NIST-OSCAL, STIG, ISO, CERT-In + unified crosswalk
- **Intel:** NVD CVE 208 golden + CPE range, CBOM 40 PQC rules, 200 golden configs, 10k fleet
- **Learning + Assets:** approved learned mappings (hot-reload) + asset inventory + 5000 risk rows



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FEASIBILITY AND VIABILITY

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Feasibility :

- **Configs + NVD + framework data** already available for us.
- **Models + parsers + APIs** built with proven open tools.
- Dry run only so live devices stay untouched and safe.
- Docker based **setup works on prem** and offline without change.
- New vendor added by mapping alone with no engine rewrite
- **Human approval keeps learning accurate** and audit ready
- **36h plan:** polish UI + add mappings + PDF core built

Commercial Feasibility :

- Strong demand from govt + PSU + enterprise networks.
- Replaces manual audit effort with faster, repeatable reports.
- **On prem licensing + support + PQC migration** services.
- Saves cost by early risk find and fleet wide reuse.
- Scales to new vendors without extra tool cost.
- **Works offline** so air gapped deals stay possible.
- Recurring value via yearly compliance + audit renewals

Challenges :

- **Vendor syntax** drift across OS and models
- Unknown lines need **human review** before use
- **AI output** must pass strict schema checks
- **CVE volume** needs constant sync and triage
- Old crypto still present in field devices
- **Large fleets** need fast dedup and scoring
- **Offline mode** limits live threat lookup
- Approval load grows with each new vendor
- **PDF proof** must stay readable for auditors

Strategy :

- **Single neutral model** + registry absorbs drift
- **Queue + suggest + approve** handles unknowns
- Schema gate blocks bad AI before audit
- Scheduled sync + range match keeps CVE exact
- **CBOM plus readiness plan** drives crypto upgrade
- **Risk score + dedup + fleet** view scales review
- Offline fallback + cache keeps audits running
- Learned reuse cuts repeat approvals to near zero
- **Short evidence + visual summary** keeps PDF clear

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IMPACT AND BENEFITS

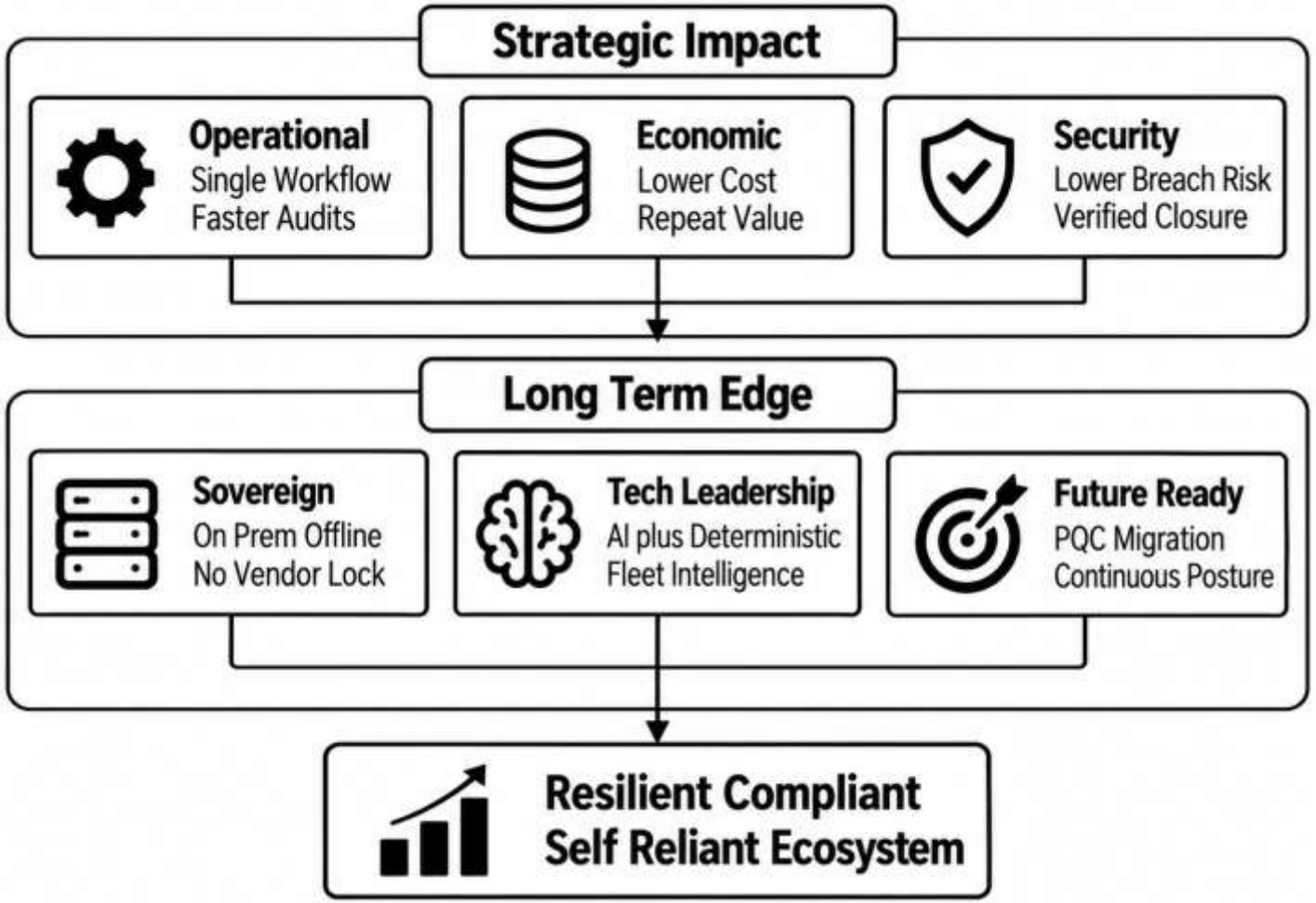
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Economic Benefits:

- **Cuts manual audit** effort to quick repeat runs
- Early **risk findings avoid costly breaches** and downtim
- One tool covers **all vendors**, so license load drops
- Fleet reuse turns one learning into many audits
- Offline use opens **govt and air-gapped** orders
- Yearly reports + rescoring give repeat revenue
- **Quantum-ready plan** avoids future rework costs
- Less expert time needed per device so team scales

Direct Impact:

- **Misconfigs caught before exploitation** with line proof
- Unknown devices become auditable after one approval
- Fleet **outliers flagged** before spread
- Fixes applied in simulation with verified closure
- **Audit posture visible** to leadership in one view
- Weak **crypto mapped to PQC** path for future safety
- **CVE blast radius** shows true exposure fast
- Compliance **score rises** across repeat audits
- Trust grows with provable evidence for every finding



Roadmap: live-SSH + scheduled NVD + full SSO

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RESEARCH AND REFERENCES

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Comparison:

Feature	Ours	Tenable	Qualys PC	SolarWinds NCM
Multi-vendor config ingest	✓	✓	✓	✓
CIS / NIST compliance checks	✓	✓	✓	✓
CVE detection	✓	✓	✓	✗
Per-device PDF report	✓	✓	✓	✓
Device-specific CLI fix	✓	✗	✗	✓
5 frameworks in single run	✓	✗	✓	✗
Unknown syntax auto-learn	✓	✗	✗	✗
PQC readiness + Q-Day	✓	✗	✗	✗
Fleet anomaly + dedup	✓	✗	✓	✗
Simulated fix + re-audit proof	✓	✗	✗	✗
Offline air-gapped full audit	✓	✗	✗	✓

References:

1. **IEEE Survey - Forensics and Compliance Auditing:** Survey of audit methods for critical infra, base for evidence-based auditing https://ieeexplore.ieee.org/document/10378648
2. **IEEE/ACM ConfigChecker - LLM + Knowledge Graph Validation:** LLM + knowledge graph for auto config validation, base for our M3/M4 + OKF design https://ieeexplore.ieee.org/document/10469536
3. **CIS Benchmarks:** Hardening guides for routers, switches, and firewalls https://www.cisecurity.org/cis-benchmarks
4. **NIST SP 800-53 Rev.5 + PQC FIPS 203/204/205:** Controls catalogue via OSCAL + quantum-safe standards for Q-Day mapping https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final
5. **DISA STIGs:** Defence checklists, strict pass/fail rules https://public.cyber.mil/stigs/
6. **ISO 27001:2022:** ISMS controls standard https://www.iso.org/standard/27001

1. **GitHub link:** https://github.com/Satish-vaishyar/Multi-Vendor-Security
2. **Video Link:** https://drive.google.com/drive/folders/1kFOQInvEfgVneLH1T5mVqNEMhsopbOY2?usp=sharing
3. **Documentation Link:** https://drive.google.com/drive/folders/1WnLyJpBmT2QIBJxoXJdHWmXa6L8iEWaU?usp=sharing